MATH3030: Coursework

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Multivariate Statistics
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1 Question 1

1.1 Exploratory Data Analysis

The UN dataset contains GDP per capita, life expectancy, and population for 141 countries at 12 evenly spaced time points from 1952 to 2007. No values are missing. Sample sizes per continent are uneven (52 African, 32 Asian, 30 European, 25 American, and 2 Oceanian), so any continent-level summary for Oceania is effectively a two-country average of Australia and New Zealand. We summarise the 70-year dynamics with continent-mean trajectories for each variable and a table of 1952 and 2007 medians.

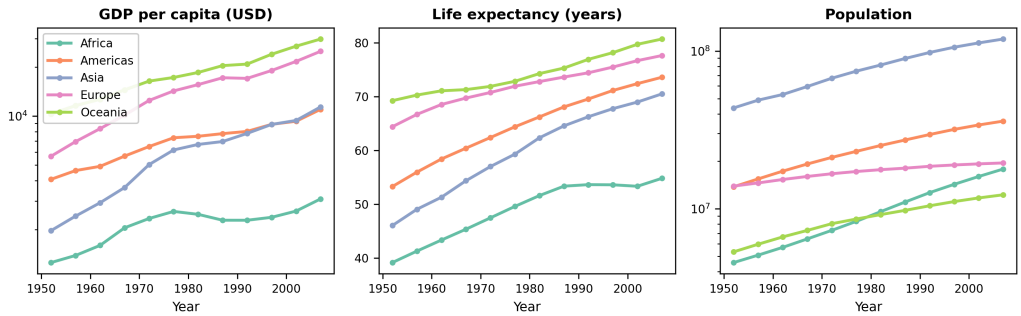

Figure 1. Continent-mean trajectories, 1952–2007. GDP and population are on log scales; life expectancy is linear.

Asia shows the steepest gains on both axes. Mean life expectancy rose from 44 to 72 years and median GDP per capita almost quadrupled, driven by a handful of export-led economies: Singapore's GDP per capita rose from \$2,315 to \$47,143 (20×), South Korea's by 23×, and Taiwan's by 24×. China dominates the population panel, rising from 556 million in 1952 to 1.32 billion in 2007. These multipliers are well above the Asian median GDP gain of 3.8×, so the continent mean hides a wide spread rather than uniform growth.

Africa's trajectory is much shallower. Median GDP rose only 1.5× over 55 years, and mean life expectancy, after rising from 39 to 53 years, plateaued from the 1990s onward. The plateau coincides with the peak of the HIV/AIDS epidemic in southern Africa and a series of regional conflicts. Zimbabwe's life expectancy fell from 48.5 to 43.5 years (the largest decline in the dataset), Swaziland, Zambia, and Lesotho show similar stagnation, and the Democratic Republic of the Congo is the only country whose GDP per capita ended below its 1952 level, at 36% of its original value. Africa's population grew faster than any other continent's, which compounds the GDP-growth shortfall on a per-capita basis.

Europe and Oceania remain firmly at the top throughout. Europe's GDP growth ($5,143 to $28,054, 5.5×) is comparable to Asia's in multiplier terms, but its life-expectancy gain of +13 years is about half of Asia's +28. This is structural rather than a lack of development: a population already close to its demographic ceiling has less room to gain years than one that is catching up. Both continents also evolve more smoothly than Asia or Africa, with no sharp deviations between adjacent five-year intervals.

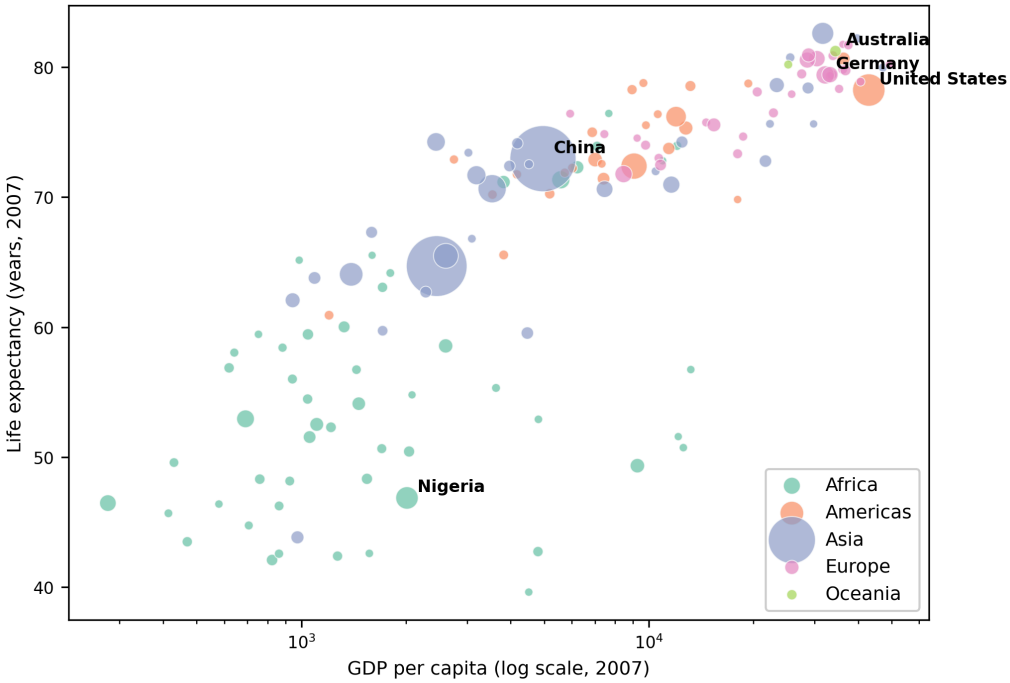

Figure 2. GDP per capita vs life expectancy in 2007. Bubble size is proportional to population.

The 2007 snapshot compresses the trajectories into a single cross-section and makes the joint structure visible. GDP and life expectancy are strongly but non-linearly related: the Pearson correlation on log(GDP) is 0.81, appreciably higher than on raw GDP (0.68), reflecting the diminishing marginal returns on life expectancy at high income. Continents separate cleanly, with Africa in the lower-left, Europe in the upper-right, and the Americas grouped in the middle. Asia is the outlier continent in this view, spanning almost the full income range from India and Bangladesh at the low end to Japan and Singapore at the top, with China positioned between the two modes.

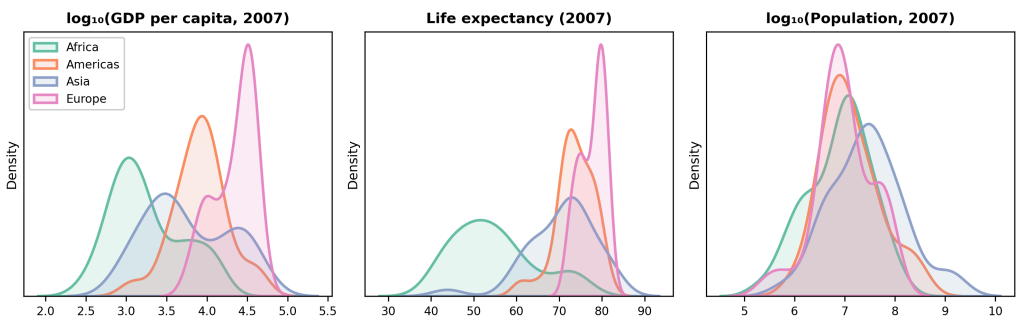

Figure 3. Kernel density estimates of each variable in 2007, one curve per continent.

The KDEs show the within-continent distributional shape. Asia's GDP distribution is visibly bimodal (Gulf states and East Asian tigers on one side, South and Central Asia on the other), with an IQR ratio

of 8.9, the largest of any continent. Africa's life-expectancy distribution is wide and left-skewed, with a lower tail formed by Swaziland, Zambia, Zimbabwe, and Lesotho, all below 45 years. Population densities peak near 10 million in every continent, with Asia carrying the long right tail that contains China and India. Continents that appeared internally homogeneous on the trajectory plot thus conceal real heterogeneity under the mean.

Continent	N	GDP 1952	GDP 2007	LE 1952	LE 2007	Pop 2007 (M)
Africa	52	987	1,452	38.8	52.9	929
Americas	25	3,048	8,948	54.7	72.9	899
Asia	32	1,148	4,328	44.4	72.2	3{,}812
Europe	30	5,143	28,054	65.9	78.6	586
Oceania	2	10,298	29,810	69.3	80.7	24

Table 1. Median GDP per capita (USD) and life expectancy (years) by continent; population (millions).

A small number of countries deviate strongly from their continent's trend and are expected to appear as extreme points in the PCA, CCA, and MDS projections below. Equatorial Guinea's GDP per capita rose 32× on the back of offshore oil discoveries; Zimbabwe's life expectancy fell by five years; Singapore, South Korea, and Taiwan achieved sustained 20–24× GDP multiples; and the Democratic Republic of the Congo and Haiti are the two economies that regressed most sharply, ending the period at 36% and 65% of their 1952 GDP per capita respectively.

1.2 Principal Component Analysis

We perform PCA independently on the 141 × 12 GDP and life-expectancy blocks. GDP values span more than two orders of magnitude across countries, so we standardise each year column, which is just decomposing the correlation matrix, to prevent larger recent-year variances from dominating. Life expectancy is on a comparable scale across years, but we also standardise it; the covariance and correlation decompositions produced near-identical results (PC1 explains 92.0% and 92.3% of the variance respectively), and standardising keeps the treatment of the two variable blocks consistent.

1.2.1 Scree analysis and variance explained

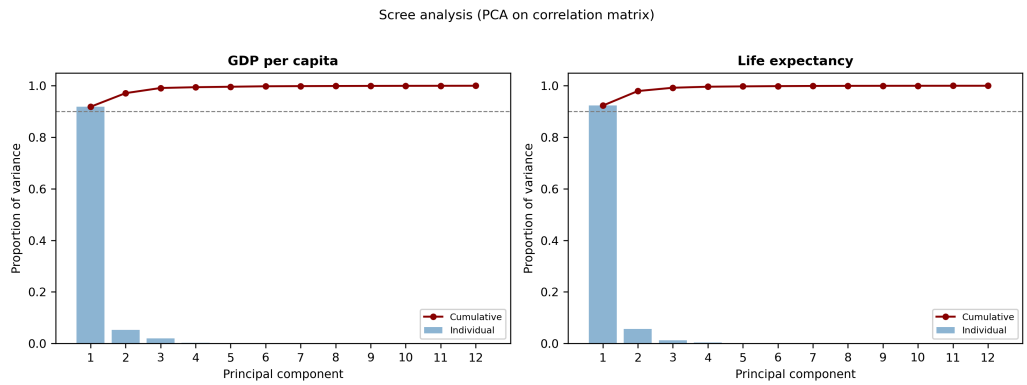

Figure 4. Scree plots for the GDP and life-expectancy PCAs. Both are dominated by a single leading component.

Both decompositions concentrate almost all variance in a single component. PC1 explains 91.9% of the GDP variance and 92.3% of the life-expectancy variance while PC2 adds a further 5.3% and 5.6% respectively. Two components are sufficient as together they cover 97.1% of the GDP variance and 97.9% of the life-expectancy variance. This follows directly from the year-to-year redundancy within

each block, with adjacent years correlated at approximately 0.99, so 12 columns compress cleanly into two.

1.2.2 Interpretation of PC1 and PC2

The loadings have the same structure on both decompositions. PC1 assigns nearly uniform positive weights across all 12 years (GDP: 0.28 to 0.30; life expectancy: 0.27 to 0.30), so a country's PC1 score is approximately proportional to its average level over the full 55-year period. Whereas for PC2, the loadings begin strongly negative in 1952 (-0.35) and rise monotonically to strongly positive in 2007 ($+0.42$). A positive PC2 score therefore indicates a country whose values were low early and high late (a catching-up or fast-growth trajectory), and a negative PC2 score indicates the opposite (stagnation or decline from a relatively high base).

1.2.3 PC score scatter plots

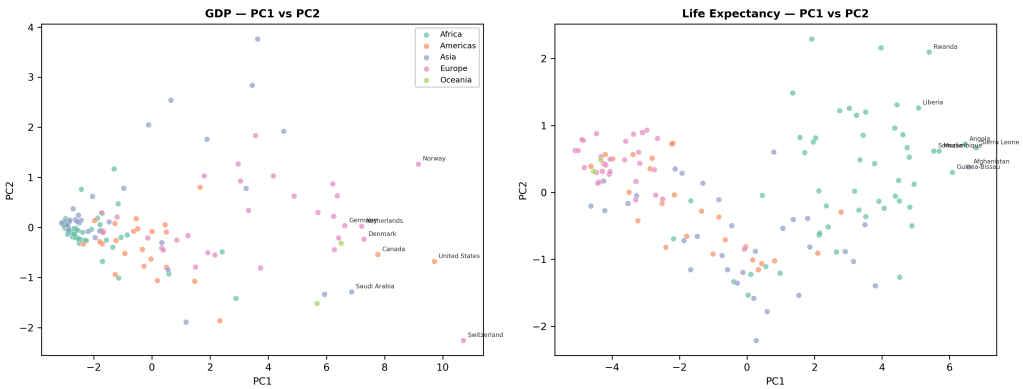

Figure 5. PC1 vs PC2 scatter plots for GDP (left) and life expectancy (right), coloured by continent.

The GDP scatter is structured by level on PC1. Switzerland (+10.7), the US (+9.7), Norway, Canada, Denmark sit at the top end while Myanmar, Burundi, Ethiopia, Eritrea, and Malawi at the bottom. PC2 then separates countries by growth trajectory. The top Asian countries form the obvious cluster at the top of PC2 (Singapore +3.8, Hong Kong +2.8, Taiwan +2.5, South Korea +2.1, Japan +1.9), which identifies the countries whose relative position improved most over the period. Strongly negative PC2 scores belong to Venezuela, Iraq, and Libya, relatively high-income economies in 1952 that did not sustain their early position advantage.

The life-expectancy scatter has the same geometry. PC1 separates Iceland, Sweden, Norway, the Netherlands, and Switzerland at the top and Sierra Leone, Afghanistan, Angola, Guinea-Bissau, and Mozambique at the bottom. PC2 captures the rate of improvement with Oman (+2.2), Vietnam, Saudi Arabia, Indonesia, and Libya gaining life expectancy fastest, while Zimbabwe (-2.3), Zambia, Rwanda, Botswana, and the Democratic Republic of the Congo have the strongly negative PC2 scores consistent with the health epidemic and regional conflicts identified in the EDA. Continent clusters are distinct with Europe and Oceania at the top of PC1, Africa at the bottom, the Americas in the middle, and Asia stretched along PC2 consistent with its bimodal GDP distribution.

1.2.4 GDP PC1 versus life-expectancy PC1

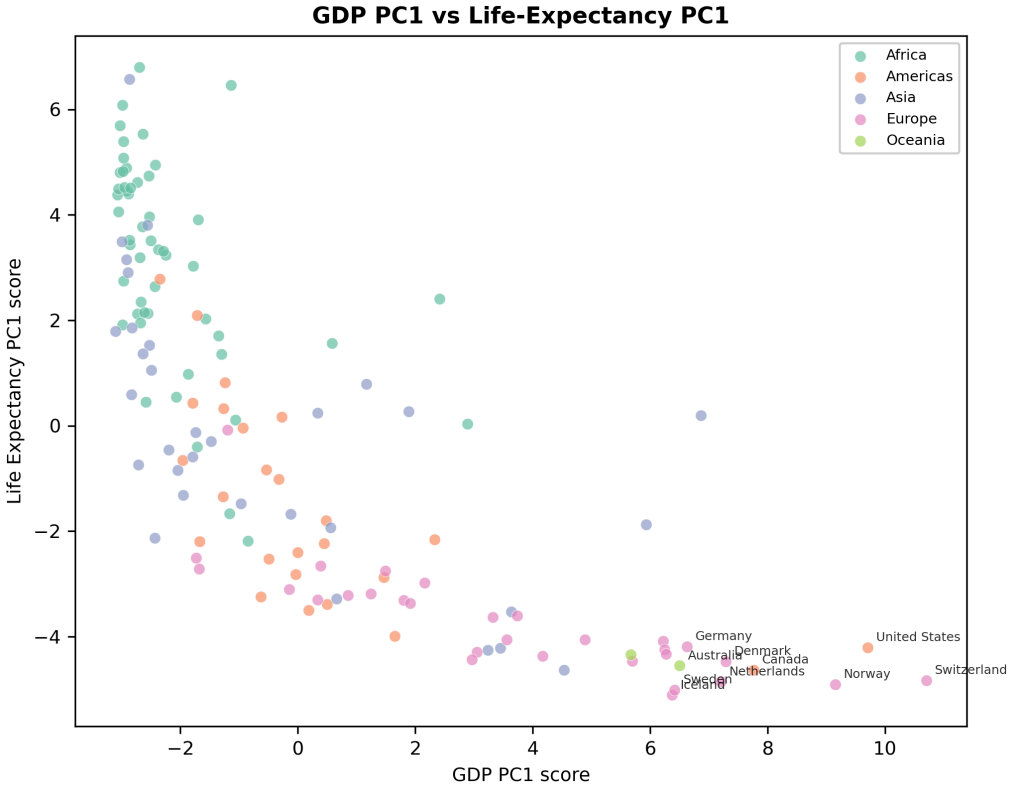

Figure 6. First PC scores, GDP against life expectancy.

Plotting the two leading PC1 scores against each other presents the relationship between overall economic level and overall health levels across the full period. The Pearson correlation is 0.77, consistent with the 0.81 log(GDP)–life expectancy correlation seen in the 2007 snapshot and confirming that wealth and health run together over long time horizons. Most countries lie close to the diagonal; the systematic deviations are countries whose health outcomes are either stronger or weaker than their income level would predict, reflecting external factors such as public health investment, diet, and health system access that the economic dimension alone does not capture.

1.2.5 Limitations of linear PCA

The decomposition is restricted to linear combinations of the year columns and the GDP–life expectancy relationship is non-linear (Pearson correlation 0.68 on raw GDP against 0.81 on log(GDP)). Kernel PCA, which applies the standard decomposition in a feature space induced by a kernel, would capture this curvature directly and would be expected to produce tighter continent separation than the linear projection used here.[1]

1.3 Canonical Correlation Analysis

We apply CCA with $\log(\mathrm{GDP})$ as the first variable set and life expectancy as the second, giving $p = q = 12$. The log transform is motivated by the same skewness and non-linearity noted in the EDA and the PCA limitations subsection, since raw GDP is strongly right-skewed (skewness 1.88) and the 2007 GDP–life expectancy relationship curves noticeably on the raw scale but is close to linear on the log scale.

[1]Background reading carried out for a separate statistical project this year; a fuller treatment is given in Hastie, Tibshirani, and Friedman, *The Elements of Statistical Learning* (2nd ed., 2009), Chapter 14.

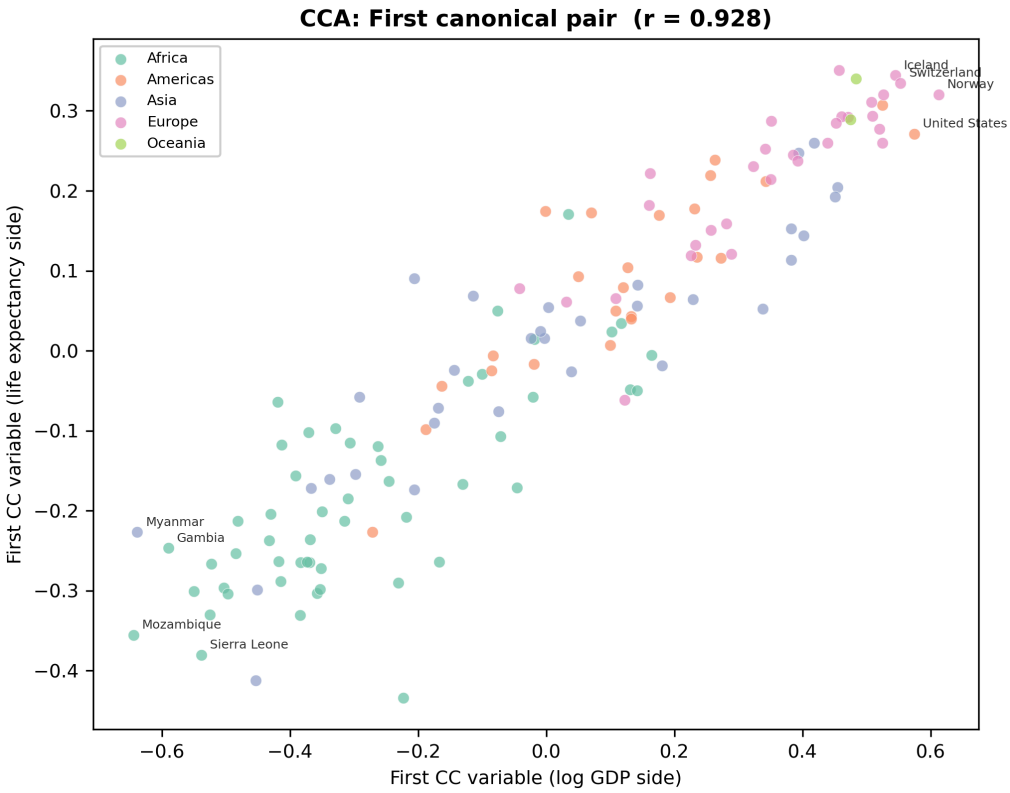

Figure 7. First pair of canonical variables, coloured by continent.

1.3.1 Canonical correlations

The first canonical correlation is 0.93, and subsequent correlations decay rapidly to 0.63, 0.54, and 0.50, with the remainder below 0.40. The first pair therefore captures most of the shared linear structure between the two blocks, and the higher-order pairs are unlikely to admit a clean interpretation given how quickly they fall off.

1.3.2 Interpretation of the first canonical pair

The first canonical axis ranks countries by a joint development level that log(GDP) and life expectancy identify together. The lower extreme is dominated by sub-Saharan African and conflict-affected Asian states, with Mozambique, Myanmar, and Sierra Leone at the bottom, and the upper extreme by Western European countries and the United States, with Norway and Switzerland at the top. This ordering is near-identical to the one PCA produced on either variable separately, because the 12 year columns within each block are correlated at approximately 0.99 and there is little room for CCA's maximum-correlation objective to choose a direction meaningfully different from PCA's maximum-variance one. The continent structure on the canonical scatter reproduces the PCA pattern, with Africa in the lower-left and Europe and Oceania in the upper-right.

1.3.3 Effect of the log transform

Repeating the analysis on raw GDP gives a first canonical correlation of 0.89, materially lower than the 0.93 achieved with log(GDP). CCA only finds linear combinations of the variables it is given, so any curvature in the raw GDP–life expectancy relationship is left unaccounted for. The log transform straightens the relationship and raises the canonical correlation, and this 0.04 gain is the direct numerical expression of the linearity limitation noted earlier in the PCA section.

1.4 Multidimensional Scaling

Classical (metric) MDS is applied to the combined 36-dimensional dataset formed by concatenating log(GDP), life expectancy, and log(population), using Euclidean distance as specified in the coursework. The log transforms on GDP and population bring them onto a roughly comparable scale with life expectancy. The Pearson correlation between pairwise distances in the original 36-D space and the 2D MDS projection is 0.997, so the two retained dimensions preserve the distance geometry of the full data almost exactly.

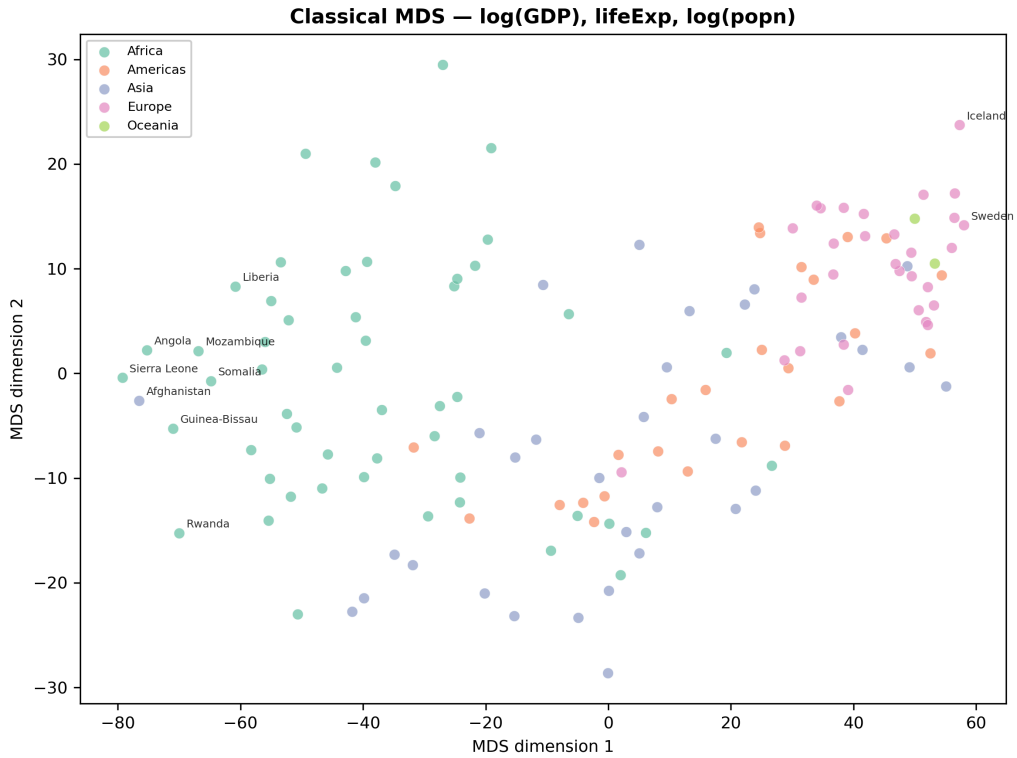

Figure 8. Two-dimensional MDS representation, coloured by continent.

1.4.1 Development axis

Dimension 1 is a clear development axis. The continent centroids line up from Africa at -37 through the Americas at $+19$ and Europe at $+43$ to Oceania at $+52$, in the same order they occupy on every previous projection. The lower extreme is dominated by sub-Saharan African and conflict-affected Asian states such as Sierra Leone, Afghanistan, Angola, Guinea-Bissau, and Rwanda, while the upper extreme is occupied by Scandinavian and Alpine European countries including Sweden, Iceland, Norway, the Netherlands, and Switzerland. This country ordering matches the one produced by PCA and CCA, since all three methods are projecting the same dominant development signal that accounts for the bulk of the variation in the data.

1.4.2 Growth direction

Dimension 2 separates countries by the direction of change in life expectancy over the period rather than by level. The most negative scores belong to countries with the largest life-expectancy gains over the 55 years, a cluster containing Oman, Vietnam, Indonesia, and Yemen, while the most positive scores belong to countries whose life expectancy fell or stagnated under the HIV epidemic, including Zimbabwe, Botswana, Zambia, and Swaziland. Asia's centroid is pulled below the horizontal axis by its concentration of fast-gaining countries, while Africa's centroid sits near the horizontal because the continent contains both rapidly improving and declining cases. The contrast is the same one

that appeared as PC2 in the life-expectancy PCA, now reproduced with all three variables analysed simultaneously.

1.4.3 Comparison with the earlier analyses

The MDS projection is consistent with the PCA and CCA results rather than revealing new structure, with dimension 1 recovering the development level axis and dimension 2 recovering the growth-direction contrast. Its distinctive contribution is that it operates on all three variables simultaneously, so a single 2D summary captures joint structure that PCA and CCA required separate analyses to recover. The 0.997 distance correlation between the original and reduced pairwise distances makes the 2D projection a near-lossless summary of the 36-dimensional dataset, which strengthens the claim that the development and growth-direction axes are the two dominant sources of variation in the full UN data.

2 Question 2

2.1 Linear Discriminant Analysis

We use LDA to predict continent from all 36 features (GDP, life expectancy, and population across 12 years). The data are split 70/30 into training and test sets. A single fixed split with seed 42 gives a test accuracy of 0.512 on 43 held-out countries, and averaging over 50 random splits gives a more stable estimate of 0.594 ± 0.064. Both figures sit comfortably above the 0.20 five-class chance rate and above the 0.37 majority-class baseline set by Africa's sample proportion, so the classifier is extracting genuine structure from the features.

True \ Pred	Africa	Americas	Asia	Europe	Oceania
Africa	11	3	2	0	0
Americas	2	3	2	1	0
Asia	3	3	2	0	0
Europe	0	2	1	6	1
Oceania	0	0	0	1	0

Table 2. Confusion matrix on the 43-country test set at seed 42. Rows are true continents, columns are predicted.

Per-class accuracy varies substantially. Africa is the best-classified continent at 0.69, followed by Europe at 0.60, the Americas at 0.38, and Asia at 0.25. The single Oceanian country in the test set is misclassified as European. Asia's errors are spread across Africa, the Americas, and Europe in roughly equal proportions. This is consistent with the EDA finding that Asia spans almost the full development range with a strongly bimodal GDP distribution. LDA assumes each class is drawn from a Gaussian with a common covariance, and a bimodal continent violates that assumption. The classifier places Asia's mean somewhere between its two modes, where it overlaps with the other continents and is frequently misclassified. Africa, the other internally heterogeneous continent, is classified more accurately because its spread is along a single unimodal axis rather than a bimodal one.

2.2 Clustering

Three methods from the module are applied to the standardised GDP and life-expectancy blocks (24 features). k-means and the Gaussian mixture model are fitted directly, and hierarchical clustering is applied via its linkage-distance dendrogram. k is chosen from the dendrogram and the GMM BIC curve, with the same k then used for k-means so that all three methods can be compared on equal footing.

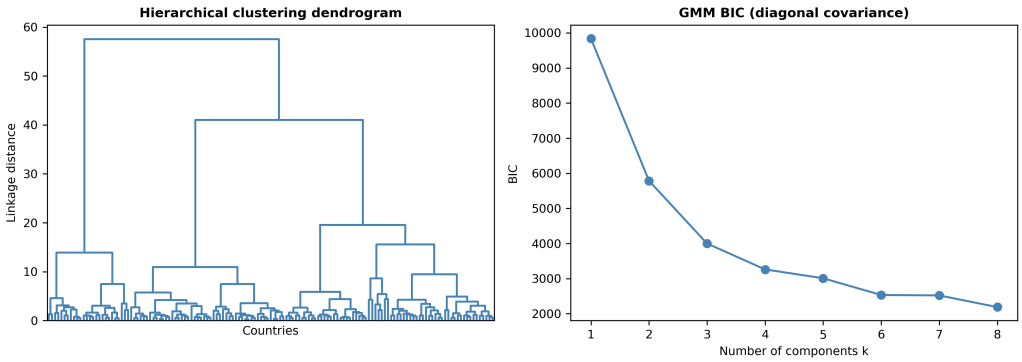

Figure 9. Left: hierarchical clustering dendrogram. Right: GMM BIC as a function of number of components.

2.2.1 Choice of k

The dendrogram's three largest merge distances are 57.6, 41.0, and 19.5, with every subsequent merge below 16. The substantial gap between merges 3 and 4 identifies $k = 3$ as the cleanest cut point. The GMM BIC curve decreases monotonically over the range we examined rather than producing a minimum. This usually indicates that the data do not form sharply separated Gaussian clumps and instead lie on a continuous development manifold, which is consistent with the MDS result showing smooth variation along dimension 1. The rate of BIC improvement slows considerably after $k = 3$, so $k = 3$ remains a defensible choice from the model-based perspective even without a true minimum.

2.2.2 Agreement between methods

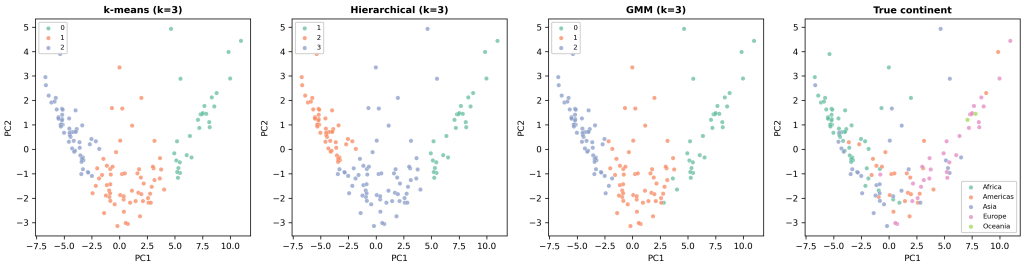

Figure 10. Cluster assignments at k = 3 for k-means, hierarchical clustering, and the GMM.

At $k = 3$ the three methods produce strongly consistent labellings. The Adjusted Rand Index is 0.84 between k-means and hierarchical clustering, 0.83 between k-means and the GMM, and 0.68 between hierarchical and the GMM. Agreement this high across the three methods is strong evidence that the cluster structure is intrinsic to the data rather than an artefact of any single algorithm.

2.2.3 Cluster interpretation

The three clusters have a natural development interpretation. The first cluster contains 18 of the 30 European countries, 6 Asian countries, and both Oceanian countries, placing it at the high-GDP and high-life-expectancy end of the development axis already seen in PCA and MDS. The second cluster contains 42 of the 52 African countries together with 10 Asian and 2 American ones, and sits at the low-development end. The third is the middle cluster, dominated by the Americas (21 of 25) with a substantial Asian contingent (16 of 32) and the remaining 12 European countries. The ordering reproduces the PCA and MDS development axis and shows that continent is a reasonable but imperfect proxy for development level. Africa and Europe fall almost entirely into the low and high clusters respectively, while the Americas and Asia are split across the middle and the extremes.

2.3 Linear Regression

We predict 2007 life expectancy from the 11 GDP columns for 1952–2002, excluding 2007 GDP to avoid contemporaneous leakage. The 11 predictors are correlated with one another at approximately 0.99 (the year-to-year structure noted in the EDA), which is textbook multicollinearity and motivates testing methods that handle it directly rather than OLS alone. Six models are compared. Ordinary least squares, ridge regression, and principal component regression (PCR) are each applied to raw GDP and to $\log(\text{GDP})$. Ridge regularisation strength is chosen by built-in cross-validation over a grid of 50 values, and PCR's number of components is chosen by cross-validation over $k = 1, \ldots, 11$. All models are scored by leave-one-out cross-validated RMSE and R^2.

Leave-one-out is appropriate for this dataset. The 141-country sample is small enough that a single 70/30 split would leave only around 99 training rows against 12 coefficients per model, and the n-fold

fitting cost is negligible for linear and ridge regression. Scoring every model on the same sequence of 141 single-country holdouts also makes the comparison between them unambiguous, because any RMSE difference between two models reflects the model specification rather than which split happened to be drawn.

2.3.1 Ridge shrinkage and principal component regression

Ridge regression minimises a penalised least-squares objective,

$$\|y - X\beta\|^2 + \alpha\|\beta\|^2,$$

which shrinks every coefficient toward zero. The amount of shrinkage applied to each direction in the predictor space is inversely proportional to the variance of that direction, so low-variance directions are penalised most. Under multicollinearity the OLS coefficients in those same directions have large variance, and ridge stabilises them at the cost of a small amount of bias. The tuning parameter α controls the trade-off, with $\alpha \to 0$ recovering OLS and $\alpha \to \infty$ driving all coefficients to zero. We select α by cross-validation over a log-sapce grid search of 50 candidates in $[10^{-2}, 10^{3}]$, taking the value that minimises the held-out prediction error. The grid is wide enough that the selected α is interior for both the raw and log specifications, so the optimum is not an artefact of the search range.

Principal component regression implements a similar idea through hard thresholding rather than continuous shrinkage. The original predictors are replaced with their first k principal components, we run OLS in the reduced space, and the low-variance PC directions are discarded entirely. This is particularly well-suited to the present setting because the PCA of the GDP block in Section 1.2 showed that two or three components already explain 97–99% of the variance, so the directions PCR discards carry negligible signal and their inclusion in OLS adds estimator variance.

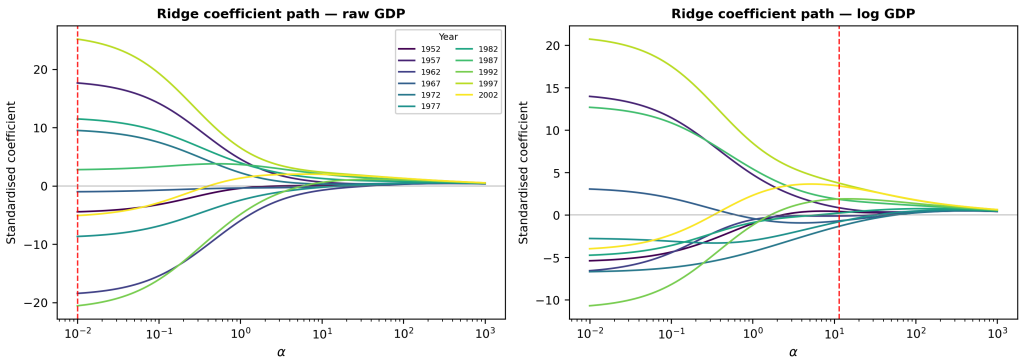

Figure 11. Standardised Ridge coefficients as a function of the penalty $alpha$, on raw GDP (left) and log GDP (right). The vertical dashed line marks the CV-optimal $alpha$ used in the main results.

At low α the eleven year coefficients are large and opposing in sign. This is expected when predictors are near-collinear, since OLS can produce many combinations of large coefficients that mostly cancel when predicting the response. Increasing α shrinks the coefficients toward zero and flattens the year-to-year differences between them. The cross-validated α sits in the intermediate region, where shrinkage has stabilised the coefficients without yet removing their signal.

Model	**LOOCV RMSE**	**LOOCV R^2**
OLS (raw GDP)	9.26	0.41
OLS (log GDP)	7.64	0.60
Ridge (raw GDP)	9.26	0.41
Ridge (log GDP)	7.21	0.64
PCR (raw GDP, $k = 2$)	8.98	0.44
PCR (log GDP, $k = 3$)	**7.15**	**0.65**

Table 3. Leave-one-out cross-validated performance for the candidate regression models.

OLS on raw GDP gives a LOOCV RMSE of 9.26 years and $R^2 = 0.41$, while OLS on log(GDP) gives RMSE 7.64 and $R^2 = 0.60$, a one-and-a-half-year reduction in prediction error from the transform alone. Ridge on log(GDP) with its penalty $\alpha = 11.5$ improves this further to RMSE 7.21 and $R^2 = 0.64$, while the penalty on raw GDP is $\alpha = 0.01$, small enough that Ridge collapses onto OLS. The mechanism is the same one discussed in the PCA and CCA sections. GDP's skewness makes the linear relationship with life expectancy non-linear on the raw scale but close to linear on the log scale, and the regression results express the cost of ignoring this as 0.19 units of R^2.

Principal component regression is the nominal winner. On log(GDP) the best configuration uses $k = 3$ components and achieves LOOCV RMSE 7.15 and $R^2 = 0.65$, narrowly ahead of Ridge. This is consistent with the PCA of the GDP block in Section 1.2, where the first three components already explain 99% of the variance, so a three-component regression retains essentially all of the signal while discarding the near-zero-variance directions that inflate the least-squares variance. PCR on raw GDP peaks at $k = 2$ with RMSE 8.98 but still does not reach the log-based models, confirming that the transform is more important than the multicollinearity fix.

The margin between PCR and Ridge warrants some caution. The number of components k is chosen by the same leave-one-out procedure that reports the final RMSE, so the tuning and evaluation overlap and the 7.15 figure is a mildly optimistic estimate of true out-of-sample error. The same concern applies to Ridge's α, which is also selected by cross-validation, so the PCR-versus-Ridge comparison is at least fair. The 0.06-year gap between them is small relative to LOOCV noise at $n = 141$, and the two methods should be treated as statistically indistinguishable at this sample size. What the data do support is that both methods provide a modest improvement over OLS on log(GDP) by treating the near-collinear GDP year columns, with no reliable basis for preferring one over the other.

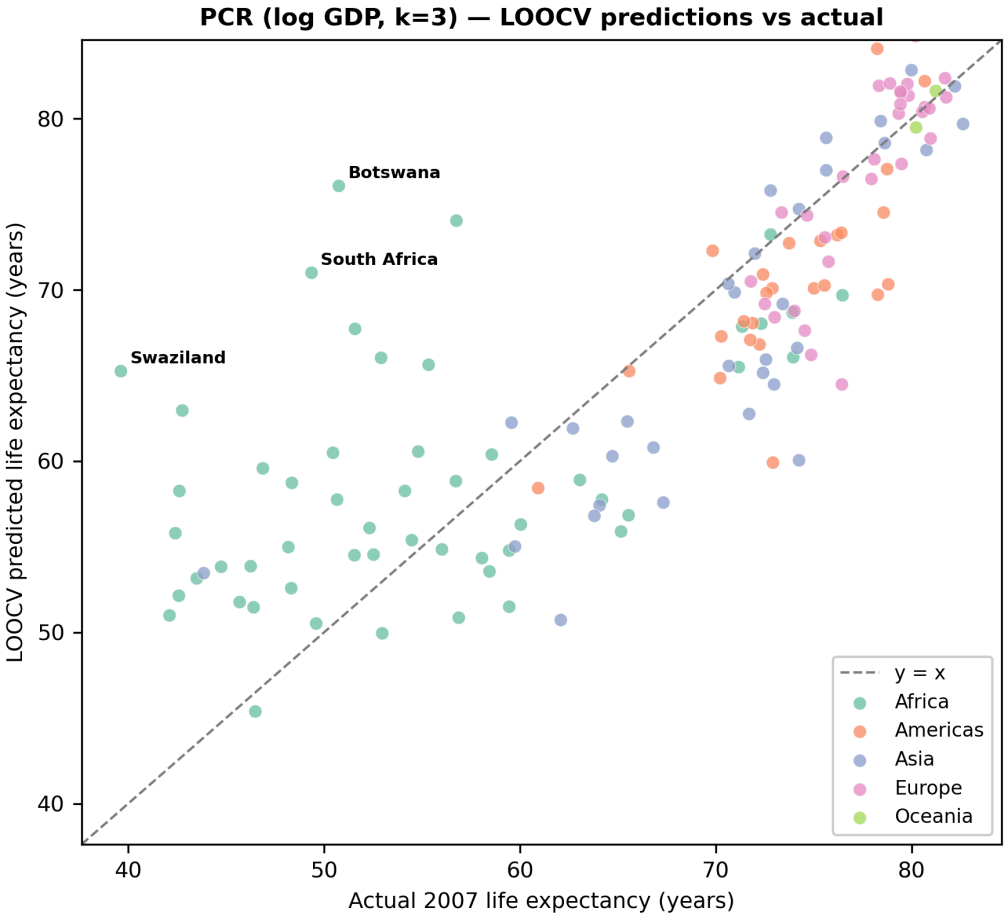

Figure 12. Leave-one-out predictions against actual 2007 life expectancy for the best model.

The largest residuals are the countries already flagged in the EDA. Zimbabwe, Swaziland, and Zambia are substantially over-predicted because their pre-1990 GDP history is consistent with the life-expectancy growth seen in most other countries, and the model has no predictor capable of capturing the HIV-driven mortality shock that followed. This is a structural limitation of using GDP alone to predict life expectancy. A health shock that leaves GDP largely unaffected is invisible to the model, and the countries in the HIV-affected cluster sit furthest from the diagonal under every combination of transform, regularisation, and component count we tried.